CHAPTER 7

WRIST MECHANISMS

7.1 INTRODUCTION

A robot manipulator needs at least 6 degrees of freedom to manipulate an object freely in space. Typically, the lengths of the first three moving links are much longer than those of the last three links. An end effector is attached to the last moving link for grasping or fine manipulation of an object. Thus the first three moving links are used primarily for manipulating the position, while the last three links are used for controlling the orientation of the end effector. For this reason, the subassembly associated with the first three moving links is called the *arm*, and the subassembly associated with the last three moving links is called the *wrist*. Furthermore, the last three joint axes are often designed to intersect at a common point called the *wrist center*. As discussed in Chapter 2, such a special arrangement leads to complete decoupling of the position from the orientation problem. The arm delivers the wrist center anywhere in its primary workspace, while the wrist controls the orientation of the end effector.

Figure 7.1 shows the kinematic structure of a 3-dof serial wrist mechanism with three intersecting joint axes, while Fig. 7.2 shows a wrist mechanism with a small offset distance between the first and second joint axes of the assembly. Theoretically, we can mount one motor with a proper gear reduction unit on each link to drive the joints. This kind of arrangement, however, requires the motors and their gear reduction units to be located close to the wrist subassembly, which will inevitably increase the inertia load to the motors of the arm subassembly. Therefore, it is highly desirable to incorporate some

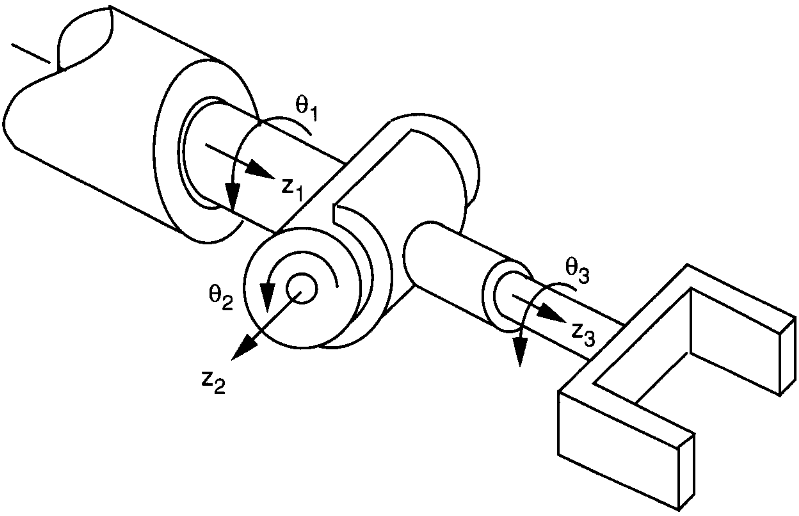


FIGURE 7.1. Spherical wrist mechanism.

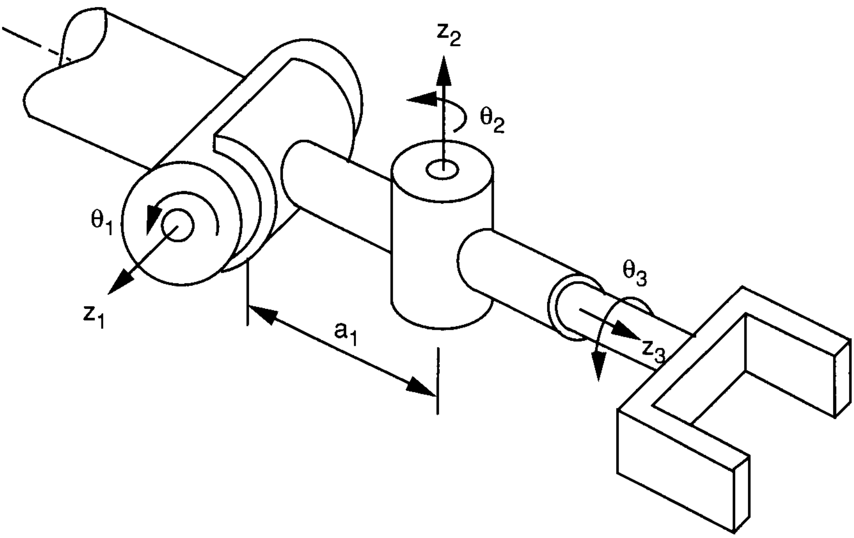


FIGURE 7.2. Nonspherical wrist mechanism.

kind of mechanical transmission mechanisms, which allow the actuators to be installed away from the wrist center.

In practice, a good wrist design should possess the following characteristics:

1. Three degrees of freedom
2. Spherical motion
3. Large workspace (i.e., large angular orientation range)
4. Remote drive capability
5. Compact size, light weight, and low inertia
6. High accuracy and repeatability
7. High mechanical stiffness
8. Low manufacturing cost
9. Rugged and reliable design

The development of wrist mechanisms can be dated back to the early nineteenth century. It is related especially to the needs in handling nuclear materials, in space exploration, and for other hazardous tasks. To achieve the necessary characteristics, mechanical transmission mechanisms such as epicyclic gear trains, push-rod linkages, and tendon drives are often employed (Rivin, 1987; Rosheim, 1989). Rosheim's book provides a thorough survey of recent developments in wrist technology, with many published and patented mechanisms, design suggestions, and other relevant information for the design of wrist mechanisms. An atlas of simple bevel-gear wrist mechanisms can be found in Lin and Tsai (1989) and Lin (1990).

Epicyclic gear drives are commonly used for speed reduction and torque amplification in mechanical systems. Bevel-gear wrist mechanisms have been incorporated in most industrial robots because they are comparatively simple and compact in size, can be sealed in a metallic housing that keeps the gear trains free of contamination, and can be produced economically and reliably. Furthermore, using bevel gear trains for power transmission, actuators can be mounted remotely on the forearm, thereby reducing the weight and inertia of a robot manipulator.

In this chapter we focus on the kinematics and statics of epicyclic gear drives and, in particular, geared robotic wrist mechanisms. First we describe some of the commonly used bevel-gear wrist mechanisms, their structure characteristics, and classifications. Then we present a systematic methodology for the kinematic analysis using the concept of fundamental circuits and the equivalent open-loop chain. Finally, we introduce the concept of transmission lines for the static analysis of such mechanisms.

7.2 BEVEL-GEAR WRIST MECHANISMS

Epicyclic gear trains have been used to transmit motion for thousands of years. The earliest known example, perhaps, is the "South-Pointing Chariot," invented around 2600 B.C. by the ancient Chinese (Dudley, 1969). This device uses an ingenious differential gear train to constrain a figurehead mounted on top of the chariot always to point to the south. It is believed that the ancient Chinese used this device to keep them from getting lost in the Gobi desert. Other early examples of gear trains include water clocks, cord-winding and rope-laying machines, and steam engines (White, 1987, 1988). A survey of planetary gear trains, complete with angular velocity equations, can be found in Glover (1964, 1965). Dudley (1969) provides a review of the historical development of gear trains from 3000 B.C. to the 1960s.

The application of epicyclic gear trains to the development of robotic mechanisms did not occur until after the nineteenth century (Anonymous, 1982; Stackhouse, 1979; Trevelyan et al., 1986). A wrist requires at least 3 degrees of freedom to locate the end effector in an arbitrary orientation. This implies a minimum of three independent rotations about three noncoplanar intersecting joint axes. If such a motion is to be realized with gearing, this necessitates a 3-dof bevel gear train of gyroscopic complexity. Furthermore, the three input links of such a gear train should be coaxial to achieve remote drive capability.

Figure 7.3 shows the schematic diagram of a wrist mechanism developed by Cincinnati-Milacron (Stackhouse, 1979). The mechanism consists of seven links (including the forearm), six turning pairs, and three bevel gear

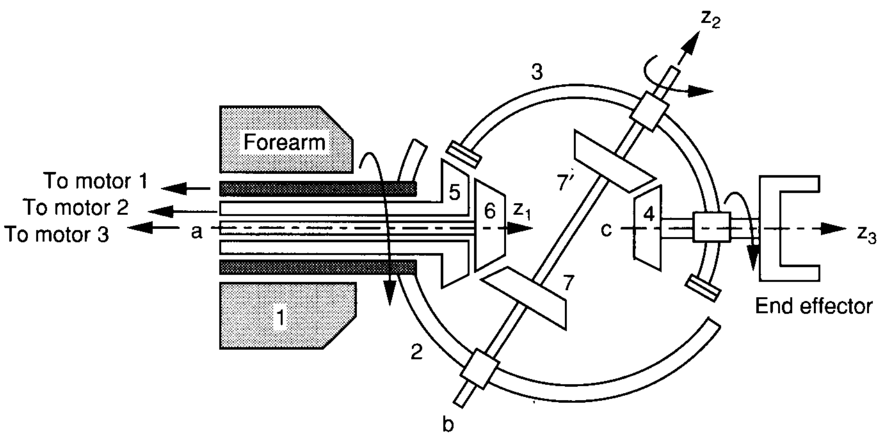


FIGURE 7.3. Cincinnati-Milacron T^3 wrist mechanism.

pairs. There are three independent axes of rotation, z_1, z_2, and z_3. The twist angle between the first and second joint axes is 60°, and between the second and third joint axes is −60°.

In the Cincinnati-Milacron wrist, link 2 serves as the carrier for the 5–3 and 6–7 bevel gear pairs, while link 3 serves as the carrier for the 7′–4 bevel gear pair. Gears 7 and 7′ are rigidly connected together by a single shaft that is housed in both carriers 2 and 3. Three coaxial members, numbered 2, 5, and 6, are supported by bearings housed in the forearm. Torques are transmitted from motors 1, 2, and 3 located in the proximal end of the forearm (not shown) to the three coaxial links. Rotation of link 5 is transmitted to link 3 via the 5–3 gear pair, while rotation of link 6 is transmitted to link 4 via the 6–7 and 7′–4 gear pairs. An end effector is attached to link 4, which is housed in the carrier 3. Furthermore, the three joint axes intersect at a common point. Therefore, the wrist mechanism is a 3-dof spherical mechanism.

The wrist roll motion is achieved by rotating link 2 with respect to link 1 about the z_1-axis. The pitch motion is accomplished by rotating link 3 with respect to link 2 about the z_2-axis. The end-effector roll motion is obtained by rotating link 4 with respect to link 3 about the z_3-axis. An important feature of this wrist mechanism is that there is no mechanical interference between all the links. Therefore, continuously unobstructed rotations about the three joint axes can be achieved.

Figure 7.4 shows the schematic diagram of another wrist developed by the Bendix Corporation (Anonymous, 1982). The Bendix wrist consists of eight links (including the forearm), seven turning pairs, and four bevel gear pairs.

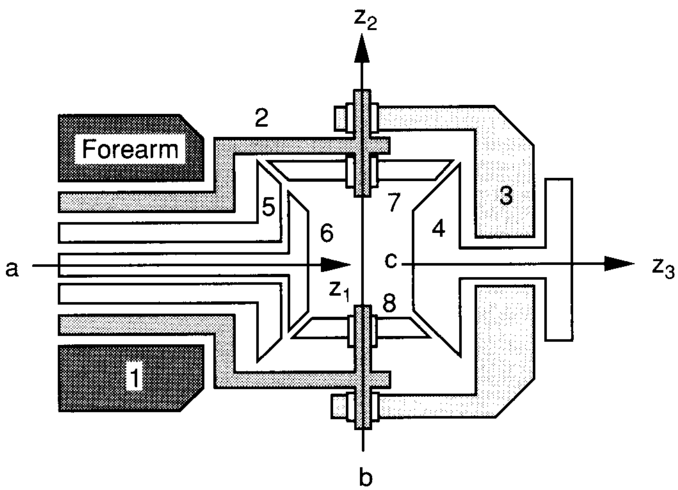


FIGURE 7.4. Bendix wrist mechanism.

There are three articulation points, axes z_1, z_2, and z_3, respectively. The twist angles are 90° between the first and second joint axes, and −90° between the second and third joint axes.

In the Bendix wrist, link 2 serves as the carrier for the 5–7 and 6–8 bevel gear pairs, while link 3 serves as the carrier for the 7–4 and 8–4 bevel gear pairs. Three coaxial members, numbered 2, 5, and 6, are supported by bearings housed in the forearm. Torques are transmitted from motors located in the proximal end of the forearm (not shown) to the three coaxial links. Bevel gear pairs 5–7, 7–4, 6–8, and 8–4 transmit rotations of the coaxial input links to the end effector attached to link 4 and housed in carrier 3. Furthermore, the three joint axes intersect at a common point. Therefore, the Bendix wrist is also a 3-dof spherical mechanism. Unlike the Cincinnati-Milacron wrist, the Bendix wrist has a limited rotational range about the second joint axis.

7.3 STRUCTURE REPRESENTATION OF MECHANISMS

In this section we describe the functional schematic, graph, and canonical graph representations of the kinematic structure of a mechanism. The following assumptions are made for all representations:

1. For simplicity, all parallel redundant paths in a mechanism will be shown as a single path only. Parallel paths are generally used to increase load capacity and to achieve a better dynamic balance of a mechanism. For example, only one planet gear will be sketched for a basic planetary gear train with multiple planets. Similarly, when a link is supported by two bearings on one shaft, only one will be shown.
2. Two mechanical elements rigidly connected for the ease of manufacturing or assembling will be considered as one link. For example, two gears keyed together with a common shaft is considered as one link from the kinematic point of view.
3. All joints are assumed to be binary. A multiple joint will be replaced by a proper combination of several binary joints. For example, a ternary joint is replaced by two coaxial binary joints, a quaternary joint is replaced by three coaxial binary joints, and so on.

7.3.1 Functional Schematic Representation

The *functional schematic representation* of a mechanism refers to the conventional drawing of the mechanism. Shafts, gears, and other elements are identified as such. For the reason of clarity and simplicity, only those elements essential to the kinematic structure are shown. For examples, the diagrams in Figs. 7.3 and 7.4 are the functional schematic representations of the Cincinnati-Milacron T^3 wrist and the Bendix wrist, respectively. In a more detailed functional schematic, link geometry and dimensions may also be shown. In this regard, two functional schematic representations of different physical embodiments may sometimes share the same kinematic structure topology.

7.3.2 Graph Representation

Since a kinematic chain is an assemblage of links and joints, this link and joint assemblage can be represented in a more abstract form known as the *graph representation.* In a graph representation, links are denoted by vertices and joints by edges. To distinguish the differences between pair connections, the edges can be colored or labeled. For epicyclic gear trains, gear pairs are denoted by heavy edges, turning pairs (revolute joints) are denoted by thin edges, and the thin edges are labeled according to their axis locations in space.

The graph of a mechanism is defined similarly, with the addition that the vertex denoting the fixed link is labeled accordingly, usually by two small concentric circles. For example, Fig. 7.5 shows the graph representation of Cincinnati-Milacron T^3 wrist. The links and corresponding vertices are numbered from 1 to 7. As shown in Fig. 7.5, the three gear pairs, 5–3, 6–7 and 7–4, are sketched in heavy edges, the turning pairs, 1–2, 2–5, 2–7, 5–6, 7–3, and 3–4, are sketched in thin edges and are labeled a, b, and c according to their axis locations.

The advantages of using graph representation can be summarized as follows:

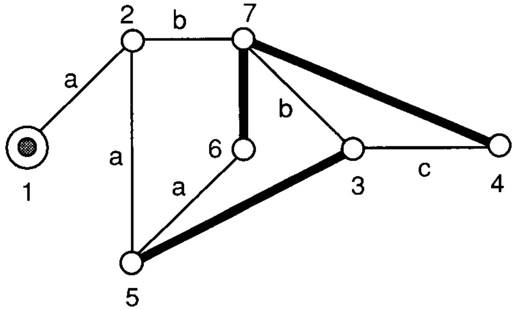


FIGURE 7.5. Graph representation of the Cincinnati-Milacron T^3 wrist.

1. Many network properties of graphs are directly applicable.
2. The structure topology of a mechanism can be uniquely identified. Using graph representation, the similarities and differences between various mechanisms can be clearly identified.
3. A single atlas of graphs can be used to enumerate an enormous amount of mechanisms (Buchsbaum and Freudenstein, 1970; Chatterjee and Tsai, 1994; Freudenstein and Maki, 1979).
4. Graphs can be used to better organize kinematic and dynamic analysis of mechanisms (Freudenstein and Yang, 1972; Roberson and Schwertassek, 1988; Tsai et al., 1996).

7.3.3 Canonical Graph Representation

When there are three or more links in a mechanism sharing a common joint axis, the turning pairs among these coaxial links can be reconfigured without affecting the functionality of the mechanism. For example, links 2, 3, and 7 shown in Fig. 7.3 are connected by two coaxial revolute joints at axis *b*. These three coaxial links can be packaged in three different ways. Figure 7.6*a* shows the original construction, in which link 2 is connected to link 7 by a revolute joint while link 7 is connected to link 3 by another revolute joint. Figure 7.6*b* and *c* shows two alternative constructions. Also shown in Fig. 7.6*a–c* are the corresponding graph representations of the three alternative constructions.

The number of alternative joint connections increases with the number of coaxial links. Mechanisms constructed from these various joint connections are called *pseudoisomorphic mechanisms*, and the corresponding graphs, *pseudoisomorphic graphs*. This suggests a canonical representation of the graph. Among all the various pseudoisomorphic graphs, the one in which all the thin-edged paths beginning from the base link and ending at any other links have distinct edge labels is called a *canonical graph*. The Cincinnati-Milacron T^3 wrist mechanism shown in Fig. 7.3 and its graph shown in Fig. 7.5 are not canonical, because the 2–7 and 7–3 edges in the 1–2–7–3–4 path are of the same label. Similarly, the 1–2, 2–5, and 5–6 edges in the 1–2–5–6 path are of the same label. After rearranging the revolute joints among the two sets of coaxial links (links 1, 2, 5, and 6) and (links 2, 3, and 7), the canonical functional schematic and graph representation of the mechanism are obtained, shown in Fig. 7.7. The canonical representation eliminates possible confusion among various pseudoisomorphic mechanisms. It leads to a systematic approach for the kinematic and dynamic analysis of geared robotic mechanisms (Tsai, 1988; Tsai et al., 1996).

In a canonical graph, the vertices can be divided into several levels. The ground-level vertex, called the *root*, denotes the base link. The first-level ver-

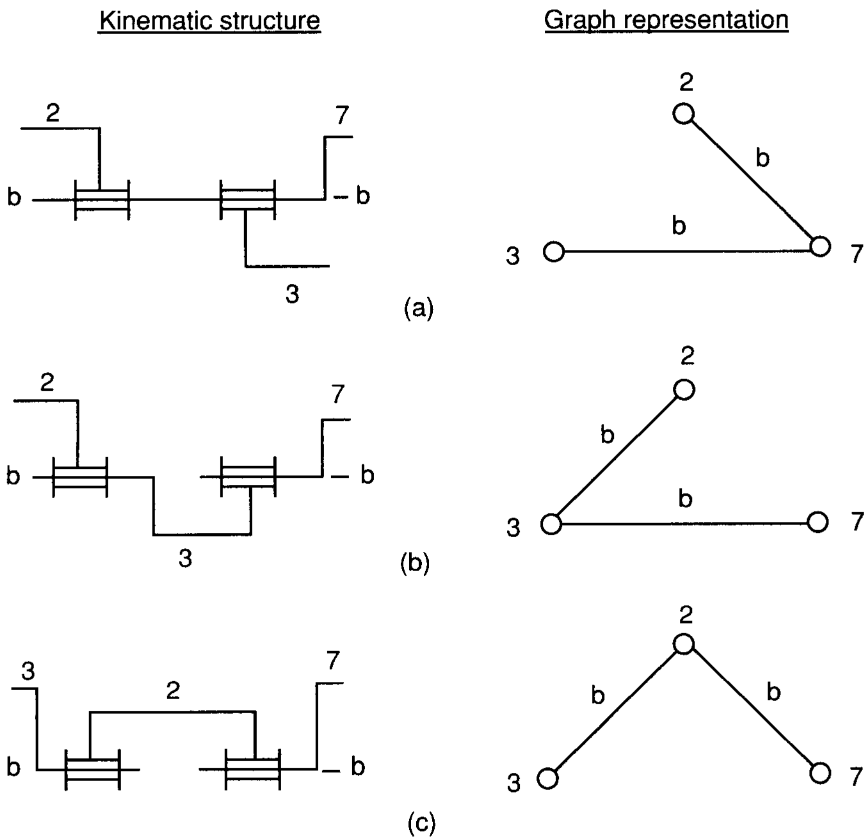


FIGURE 7.6. Three different ways of assembling three coaxial links.

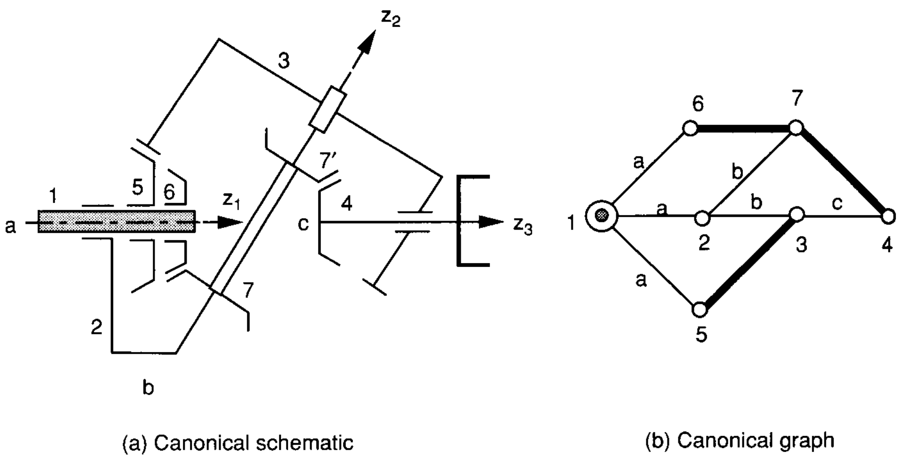


FIGURE 7.7. Canonical functional schematic and graph representation of the Cincinnati-Milacron T^3 wrist.

tices denote those links that are connected directly to the base link by a turning pair. The first-level links can be potentially employed as the input links. The higher-level vertices denote revolving planet gears or carriers, including the end effector.

7.4 STRUCTURE CHARACTERISTICS OF EPICYCLIC GEAR TRAINS

In this section we describe the structure characteristics of epicyclic gear trains. To distinguish a geared linkage from an epicyclic gear train, we impose the following rules on the kinematic structure of all epicyclic gear trains:

1. An epicyclic gear train should obey the general degree-of-freedom equation. That is, no special link lengths are required to achieve a desired mobility.
2. There are no partially locked kinematic chains within a gear train.
3. Neglecting the mechanical limit, all links in an epicyclic gear train will have unlimited rotation capability. This implies that there should be no circuits formed exclusively by turning pairs.
4. Each gear in an epicyclic gear train has a turning pair on its axis to maintain a constant center distance between a gear pair.

Let n, j, j_t, j_g, f_i, F, and L denote the number of links, the total number of joints, the number of turning pairs, the number of gear pairs, the degrees of freedom associated with the ith joint, the degrees of freedom of the gear train, and the number of independent loops, respectively. The degree-of-freedom equation can be written

$$F = 3(n - j - 1) + \sum f_i. \tag{7.1}$$

Assuming that only gear pairs and turning pairs are used in an epicyclic gear train, it is obvious that

$$j = j_t + j_g. \tag{7.2}$$

Hence the total joint degrees of freedom is

$$\sum f_i = j_t + 2j_g. \tag{7.3}$$

Rules 3 and 4 imply that the subgraph formed by removing all the geared edges from the graph of an epicyclic gear train is a tree. It is well known from graph theory that a tree of n vertices contains $n - 1$ edges:

$$j_t = n - 1. \tag{7.4}$$

Substituting Eq. (7.4) into (7.2) and (7.3) and the resulting equations into (7.1) yields

$$j_g = n - 1 - F. \tag{7.5}$$

Applying Euler's equation, we obtain

$$L = n - 1 - F = j_g. \tag{7.6}$$

From the discussions above, we summarize the structure characteristics of epicyclic gear trains as follows (Buchsbaum and Freudenstein, 1970):

1. The subgraph obtained by deleting all the geared edges from the graph of an epicyclic gear train is a tree.
2. The number of turning pairs is equal to the number of links diminished by one, $j_t = n - 1$.
3. The number of gear pairs is given by $j_g = n - 1 - F$.
4. Any geared edge added onto the tree forms one and only one circuit, called the *fundamental circuit*, or *f-circuit*. Each f-circuit contains one geared edge and several turning-pair edges.
5. The number of f-circuits is equal to the number of gear pairs, $L = j_g$.
6. Each turning-pair edge can be characterized by a label that identifies the location of its axis in space.
7. The differential degrees of freedom of any circuit must be no less than 1.
8. In each f-circuit, there is a vertex called the *transfer vertex* such that all thin edges on one side of the transfer vertex share the same label, and edges on the opposite side of the transfer vertex share a different label. The transfer vertex corresponds to the *carrier* of a gear pair.
9. Thin edges of the same label form a tree.

7.5 CLASSIFICATION OF WRIST MECHANISMS

Wrist mechanisms can be classified in several different ways. They can be classified by their degrees of freedom, type of motion (i.e., spherical versus nonspherical mechanisms), or other geometric considerations. In general, the joint axes of a wrist mechanism do not necessarily have to intersect at a common point, and the twist angles between adjacent joint axes are not necessarily equal to ±90°. A wrist is said to be a *spherical wrist* if its joint axes intersect at a common point. Otherwise, it is called a *nonspherical wrist*. A *simple* wrist is one in which the twist angles between adjacent joint axes are all equal to ±90°. On the other hand, if any of the twist angles is not equal to ±90°, it is called an *oblique wrist*. Following the definitions above, we observe that both the Cincinnati-Milacron T^3 and Bendix wrists, shown in Figs. 7.3 and 7.4, respectively, are 3-dof spherical wrists. However, the Cincinnati-Milacron T^3 wrist is an oblique wrist, whereas the Bendix wrist is a simple wrist.

Wrist mechanisms can also be classified according to their gearing arrangement. A wrist is called a *basic mechanism* if rotations of the input links are transmitted to the articulation points by gears mounted only on its axes of articulation, and it is called a *derived mechanism* if additional idler gears are incorporated between the axes of articulation points. Based on this definition and the concept of transmission lines, Chang and Tsai (1989) developed a systematic methodology for the enumeration and classification of robotic mechanisms.

Figure 7.8 shows an atlas of basic 3-dof spherical wrist mechanisms with up to nine links. Figure 7.9 shows three wrist mechanisms derived from the seven-link wrist shown in Fig. 7.8*a*. In all cases, link 1 denotes the base link (forearm) and link 4 denotes the end effector. In Fig. 7.8*a* and *b* and Fig. 7.9*a*–*c*, links 2, 5, and 6 serve as the input links, while in Fig. 7.8*c*–*e*, links 5, 6, and 7 are the input links. We note that all the mechanisms shown in Figs. 7.8 and 7.9 are sketched in simple wrist configurations. These mechanisms can easily be reconfigured into oblique wrists. Many more derived wrist mechanisms can be found in Lin and Tsai (1989) and Lin (1990).

7.6 KINEMATICS OF EPICYCLIC GEAR DRIVES

In the design of power transmission mechanisms such as speed reducers or automotive transmissions, it is often necessary to analyze the speed ratios between their input and output members and, sometimes, angular velocities of the intermediate members. The analysis is usually accomplished on a one-by-one basis using various methods, such as the tabular method (Merritt, 1947; Shigley and Uicker, 1980), the relative velocity method (Glover, 1964, 1965; Levai, 1968), the energy method (Wilkinson, 1960), the signal flow method (Ma and Gupta, 1989; Wojnarowski and Lidwin, 1973), the bond graph method (Allen, 1979), and the velocity vector-loop approach (Smith, 1979; Willis, 1982). The amount of work involved in analyzing a compound epicyclic gear train is quite laborious and can often introduce human errors. Therefore, it is highly desirable to develop a systematic procedure for the

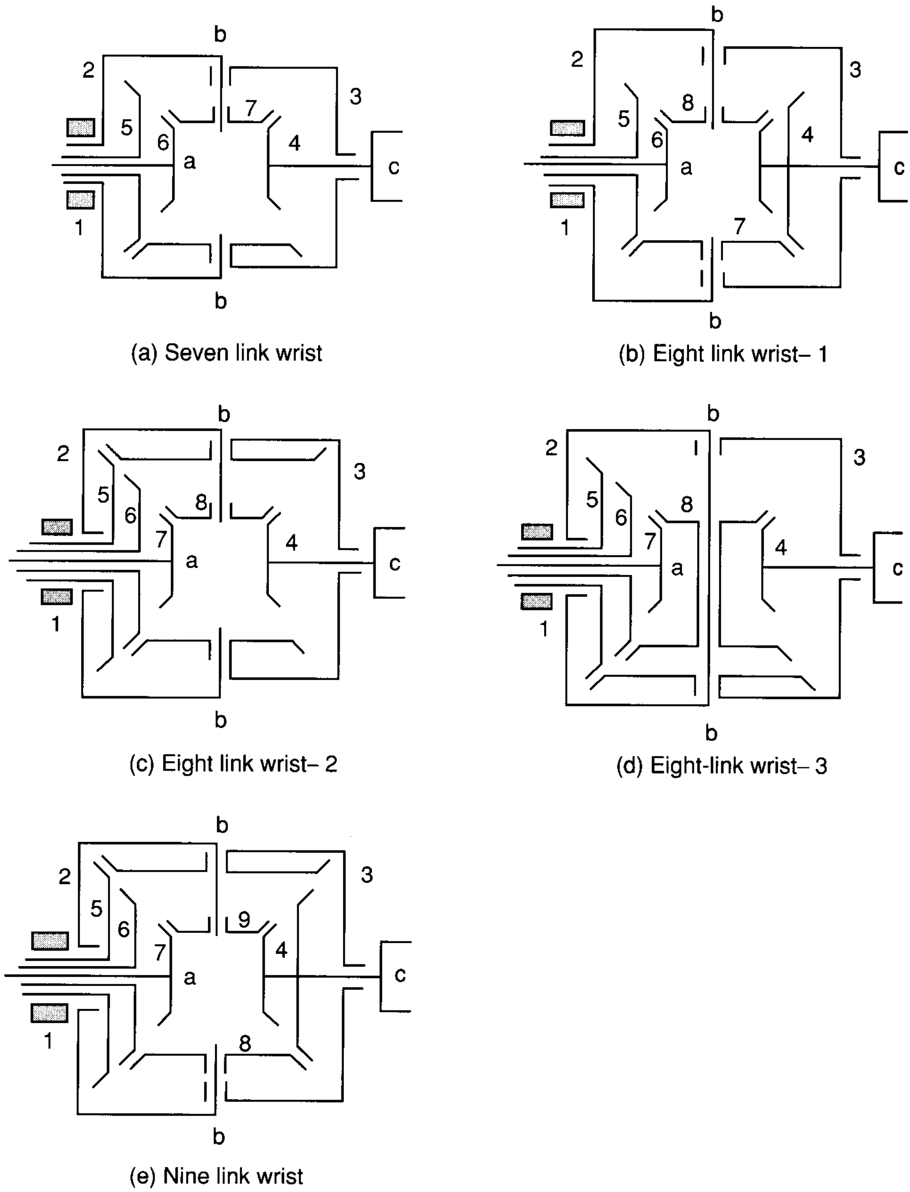
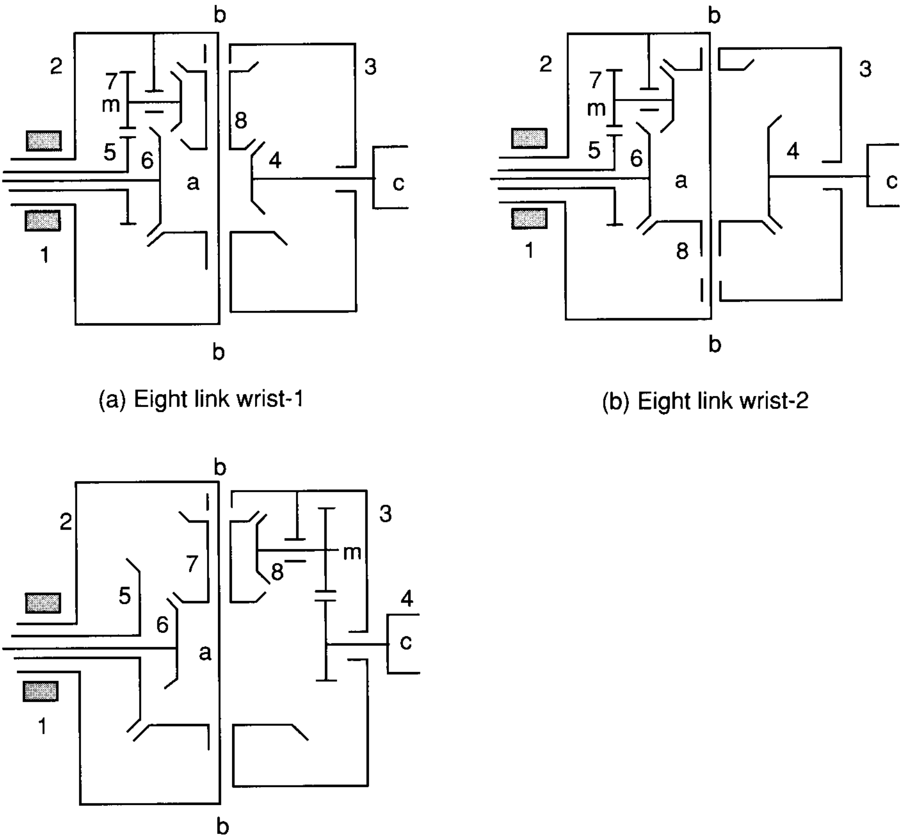


FIGURE 7.8. Five basic 3-dof wrist mechanisms.



(a) Eight link wrist-1

(b) Eight link wrist-2

(c) Eight link wrist - 3

FIGURE 7.9. Eight-link wrists derived form the seven-link wrist.

analysis (Hedman, 1989, 1993; Tsai, 1985). Perhaps the most promising approach is the systematic method introduced by Freudenstein and Yang (1972), which utilizes the theory of fundamental circuits. These workers showed that there exists a carrier for every gear pair in an epicyclic gear train and that a fundamental circuit equation can be written for each circuit. This results in a system of linear equations that can be solved for angular velocities of all the links. The method is very straightforward and can be implemented on a computer for automated analysis of epicyclic gear trains. The theory of fundamental circuits was further extended by other researchers (Belfiore and Pennestri, 1989; Pennestri and Freudenstein, 1993; Tsai, 1988).

In what follows, we introduce the theory of fundamental circuits and coaxiality condition. Then we analyze two industrial epicyclic gear drives to illustrate the theory.

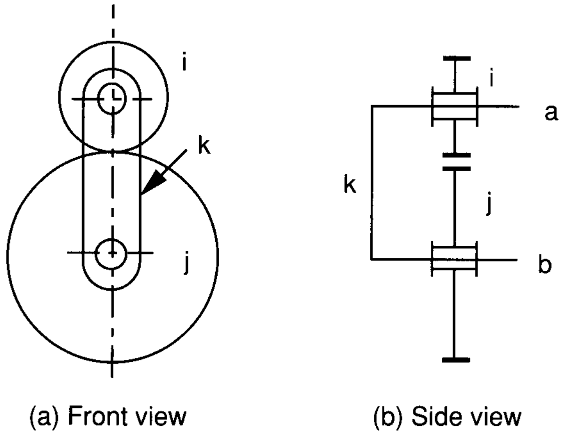


FIGURE 7.10. Gear pair.

7.6.1 Fundamental Circuit Equation

Figure 7.10 shows a meshing gear pair, i and j, in which k is the carrier. Links i, j, and k constitute a *fundamental circuit.* A *fundamental circuit equation* relating the angular displacements of the gears can be written as

$$\theta_{i,k} = \pm N_{ji}\theta_{j,k}, \tag{7.7}$$

where $\theta_{i,k}$ and $\theta_{j,k}$ denote the angular displacements of gears i and j relative to the carrier k, respectively. The gear ratio of a gear pair is defined as $N_{ji} = T_j/T_i$, where T_i and T_j denote the number of teeth on gears i and j, respectively. Following the definitions above, we have $\theta_{i,k} = -\theta_{k,i}$, and $N_{ij} = 1/N_{ji}$.

The sign in Eq. (7.7) is positive or negative depending on whether a positive rotation of gear i relative to the carrier k produces a positive or negative rotation of gear j. For planar gear trains it is convenient to assume that all the positive axes of rotation are pointing out of the paper. This results in a positive gear ratio for an internal gear mesh and a negative gear ratio for an external gear mesh. For spherical or spatial mechanisms, we assign a positive direction of rotation to each joint axis in accordance with the Denavit–Hartenberg convention. The sign of a gear ratio then follows the definition above.

Since Eq. (7.7) is written in terms of relative angular displacements, it is valid whether the gear train is planar or bevel gear type and whether the carrier is stationary or revolving. We can write one such equation for each f-circuit in an epicyclic gear mechanism, resulting in $n - F - 1$ fundamental circuit equations. Differentiating Eq. (7.7) with respect to time, we obtain an equation relating the angular velocities of the links as follows:

$$\omega_{i,k} = \pm N_{ji}\omega_{j,k}. \tag{7.8}$$

7.6.2 Coaxiality Condition

Let i, j, and k be three coaxial links; the relative angular displacements among these coaxial links can be related by the following chain rule:

$$\theta_{i,j} = \theta_{i,k} - \theta_{j,k}. \tag{7.9}$$

Equation (7.9) is called the *coaxiality condition.* The fundamental circuit equation and the coaxiality condition together can be used to solve the kinematics of epicyclic gear trains. In what follows we analyze the speed ratios of two planar epicyclic gear trains to illustrate the theory.

Example 7.6.1 *Minuteman Cover Drive* Figure 7.11*a* shows the functional schematic of the Minuteman cover drive. In this mechanism, ring gear 1 is fixed to the ground, sun gear 2 is the input link, while the other ring gear, 5, serves as the output member. The compound planet gear meshes with sun gear 2 as well as the two ring gears, and is supported by the carrier, 3, with a revolute joint. Link 2 is connected to links 1, 3, and 5 by coaxial revolute joints. Together, it forms a five-link, 1-dof compound planetary gear train. We wish to find the overall speed reduction ratio of this mechanism.

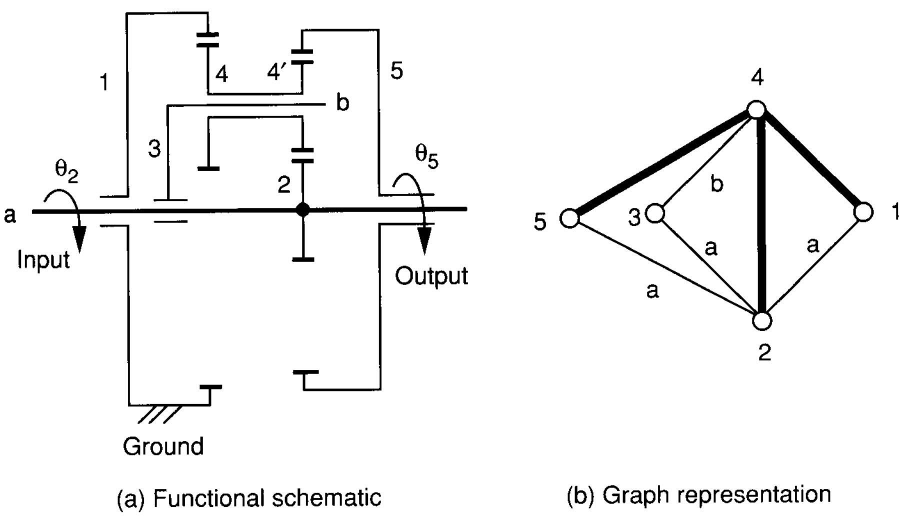


FIGURE 7.11. Minuteman cover drive and its graph representation.

Fundamental circuits. Since there are three gear pairs in the mechanism, the number of fundamental circuits is three. The graph representation of the mechanism is depicted in Fig. 7.11*b*. From the graph representation, the fundamental circuits can be easily identified. First, we remove all the heavy edges from the graph. This results in a tree (characteristic 1). Then we replace one heavy edge onto the tree at a time. According to the 4th characteristic, each time we replace a heavy edge onto the tree, we obtain a unique fundamental circuit. Finally, the transfer vertex or the carrier of each circuit is identified by applying the 8th characteristic. As a result, we obtain three fundamental circuits, (2,4,3), (1,4,3), and (4,5,3), where the first two numbers in the parentheses denote the gear pair, and the third number denotes the carrier. Hence the fundamental circuit equations can be written

$$f(2,4,3): \quad \theta_{2,3} = -N_{42}\theta_{4,3}, \tag{7.10}$$

$$f(1,4,3): \quad \theta_{1,3} = N_{41}\theta_{4,3}, \tag{7.11}$$

$$f(4,5,3): \quad \theta_{4,3} = N_{54}\theta_{5,3}. \tag{7.12}$$

Since link 1 is fixed, all angular displacements should be referred to link 1. That is, the input and output angular displacements can be written as $\theta_{2,1}$ and $\theta_{5,1}$, respectively. To derive an equation relating the output angular displacement to the input angular displacement, $\theta_{4,3}$ and $\theta_{1,3}$ should be eliminated, while $\theta_{2,3}$ and $\theta_{5,3}$ should be expressed in terms of $\theta_{2,1}$ and $\theta_{5,1}$ by applying the coaxiality conditions.

Coaxiality conditions. From Fig. 7.11 we observe that links 1, 2, 3, and 5 share a common joint axis. Two coaxiality conditions can be written as

$$\theta_{2,3} = \theta_{2,1} - \theta_{3,1}, \tag{7.13}$$

$$\theta_{5,3} = \theta_{5,1} - \theta_{3,1}. \tag{7.14}$$

Substituting (7.13) and (7.14) into (7.10) through (7.12) yields

$$\theta_{2,1} - \theta_{3,1} = -N_{42}\theta_{4,3}, \tag{7.15}$$

$$-\theta_{3,1} = N_{41}\theta_{4,3}, \tag{7.16}$$

$$\theta_{4,3} = N_{54}(\theta_{5,1} - \theta_{3,1}). \tag{7.17}$$

Equations (7.15), (7.16), and (7.17) represent a system of three linear equations in three unknowns: $\theta_{5,1}$, $\theta_{3,1}$, and $\theta_{4,3}$. Solving Eq. (7.16) for $\theta_{4,3}$ and substituting the resulting expression into Eqs. (7.15) and (7.17) yields

$$\theta_{2,1} - \theta_{3,1} - \frac{N_{42}}{N_{41}}\theta_{3,1} = 0, \tag{7.18}$$

$$\frac{1}{N_{41}}\theta_{3,1} + N_{54}(\theta_{5,1} - \theta_{3,1}) = 0. \tag{7.19}$$

Solving (7.18) for $\theta_{3,1}$ and substituting the resulting equation into (7.19) yields

$$\theta_{5,1} = \frac{N_{41}N_{54} - 1}{N_{54}(N_{41} + N_{42})}\theta_{2,1}. \tag{7.20}$$

Taking the derivative of Eq. (7.20) with respect to time and rearranging, we obtain the speed reduction ratio of the gear train as

$$\frac{\omega_{2,1}}{\omega_{5,1}} = \frac{N_{54}(N_{41} + N_{42})}{N_{41}N_{54} - 1}. \tag{7.21}$$

This compound planetary gear train can provide a very large speed reduction. For example, for $N_{41} = \frac{32}{74}$, $N_{42} = \frac{33}{9}$, and $N_{54} = \frac{75}{33}$, the speed reduction ratio is given by

$$\frac{\omega_{2,1}}{\omega_{5,1}} = \frac{\frac{75}{33}\left(\frac{32}{74} + \frac{33}{9}\right)}{\frac{32}{74} \cdot \frac{75}{33} - 1} = -541.7.$$

Example 7.6.2 *Differential Speed Reducer* Figure 7.12 shows the functional schematic of a planetary differential gear train. The mechanism uses its carrier, 2, as the input link, the larger sun gear, 3, as the output link, and the smaller sun gear, 1, as the fixed link. The compound planet gear, 4, meshes

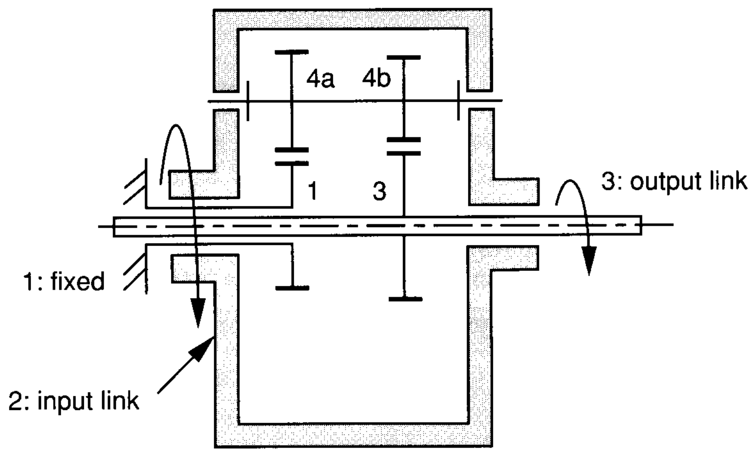


FIGURE 7.12. Differential speed reducer.

with both sun gears 1 and 3. We wish to find the speed ratio of the input link to the output link.

Since there are two gear pairs supported by a single carrier, two fundamental circuits are easily identified as (1,4,2) and (3,4,2). The fundamental circuit equations can be written

$$f(1,4,2): \quad \theta_{1,2} = -N_{41}\theta_{4,2}, \tag{7.22}$$

$$f(3,4,2): \quad \theta_{3,2} = -N_{43}\theta_{4,2}. \tag{7.23}$$

Since link 1 is fixed, all angular displacements should be referred to link 1. The input and output angular displacements can be written as $\theta_{2,1}$ and $\theta_{3,1}$, respectively. To derive an equation relating the output angular displacement to the input angular displacement, $\theta_{4,2}$ should be eliminated, while $\theta_{3,2}$ should be expressed in terms of $\theta_{2,1}$ and $\theta_{3,1}$ by applying the coaxiality condition. From Fig. 7.12 we observe that links 1, 2, and 3 share a common joint axis. A coaxiality condition can be written as

$$\theta_{3,2} = \theta_{3,1} - \theta_{2,1}. \tag{7.24}$$

Dividing Eq. (7.22) by (7.23) yields

$$-N_{43}\theta_{2,1} = N_{41}\theta_{3,2}. \tag{7.25}$$

Substituting Eq. (7.24) into (7.25) and simplifying yields

$$\frac{\theta_{2,1}}{\theta_{3,1}} = \frac{N_{41}}{N_{41} - N_{43}}. \tag{7.26}$$

Taking the derivative of Eq. (7.26) with respect to time and rearranging, we obtain the speed reduction ratio of the gear train as

$$\frac{\omega_{2,1}}{\omega_{3,1}} = \frac{N_{41}}{N_{41} - N_{43}}. \tag{7.27}$$

For example, let $N_{41} = \frac{33}{75}$ and $N_{43} = \frac{32}{76}$; then the speed reduction ratio is given by

$$\frac{\omega_{2,1}}{\omega_{3,1}} = \frac{\frac{33}{75}}{\frac{33}{75} - \frac{32}{76}} = 23.2.$$

7.7 KINEMATICS OF ROBOTIC WRIST MECHANISMS

In this section we study the kinematics of robotic wrist mechanisms of gyroscopic complexity. The analysis of bevel-gear wrist mechanisms is much more complex, due to the fact that the carriers and planet gears may possess simultaneous angular velocities about nonparallel axes. The conventional tabular or analytical method in textbooks, which concentrates on planar epicyclic gear trains, is no longer applicable. To overcome this difficulty, Yang and Freudenstein (1973) applied the dual relative velocity and dual matrix of transformation for the analysis of epicyclic bevel-gear trains and hypoid gears. Freudenstein et al. (1984) suggested using the conventional tabular method in conjunction with the Rodrigues equation. Recently, Tsai (1988) showed that the kinematic analysis of geared robotic mechanisms can be accomplished by applying the concept of *equivalent open-loop chain* and the theory of fundamental circuits. This method, which appears to be the simplest and most systematic approach, is the main focus of this section.

The method consists of two basic steps. The first step involves the identification of an equivalent open-loop chain and the derivation of the kinematic relationship between the orientation of the end effector and the joint angles of the equivalent open-loop chain. The second step involves the derivation of the kinematic relationship between the joint angles and the input actuator displacements. The first step results in a transformation between the end-effector space and the joint space of the equivalent open-loop chain, and the second step results in a transformation between the the joint space and the input actuator space. The overall kinematic relation is obtained as a product of the two transformations.

In what follows, we use the Cincinnati-Milacron T^3 wrist to illustrate the procedure.

7.7.1 Equivalent Open-Loop Chain

Recall that the graph of a geared robotic mechanism can always be transformed into a canonical form, and the removal of all the geared edges from the graph results in a tree. The tree represents an open-loop kinematic chain made up of links connected together by revolute pairs. In particular, we call the thin-edged path that starts from the base link and ends at the end-effector link the *equivalent open-loop chain*. For example, the canonical graph of the Cincinnati-Milacron T^3 wrist shown in Fig. 7.7*b* contains vertex 1 as the base link and 4 as the end-effector. The thin-edged path that starts from the base link and ends at the end-effector link consists of vertices 1–2–3–4. Hence the equivalent open-loop chain is made up of links 1, 2, 3, and 4 connected in series by revolute joints as shown in Fig. 7.13.

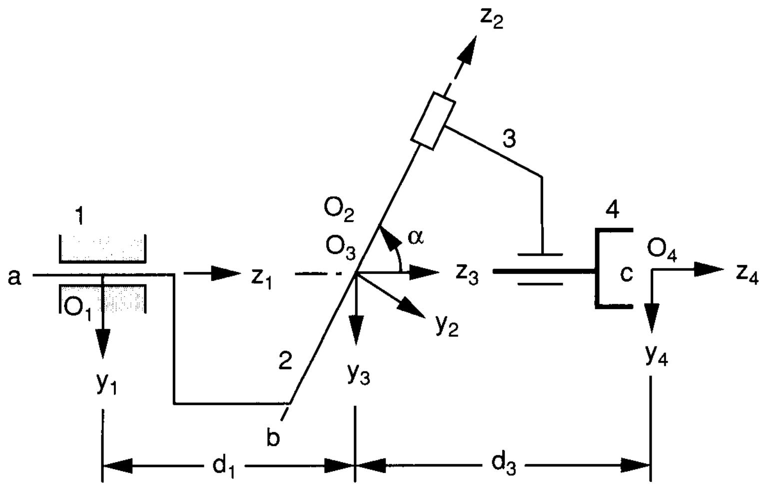


FIGURE 7.13. Equivalent open-loop chain of the Cincinnati-Milacron T[3] wrist.

To facilitate the analysis of such a mechanism, we first assign a positive direction of rotation to each joint axis and define a coordinate system for each link in accordance with the Denavit–Hartenberg convention. Thus the (x_1, y_1, z_1) coordinate frame is attached to link 1, the (x_2, y_2, z_2) coordinate frame is attached to link 2, the (x_3, y_3, z_3) coordinate frame is attached to link 3, and the (x_4, y_4, z_4) coordinate frame is attached to link 4. The origin O_1 and the x_1-axis have been chosen arbitrarily since at this time we do not know how the wrist is attached to the forearm. The equivalent open-loop chain shown in Fig. 7.13 has been sketched in a configuration in which all the x_i, $i = 1$ to 4, axes are parallel to each other and are pointing out of the paper. We define this particular configuration as the *reference position* and refer the angular displacement of a link (or gear) in the mechanism as the rotation of the link from this reference position, positive or negative in accordance with the right-hand screw rule. The Denavit–Hartenberg parameters for this equivalent open-loop chain are given in Table 7.1.

The transformation between the end-effector space and the joint space of the equivalent open-loop chain can be derived by applying the Denavit–

TABLE 7.1. Link Parameters of the Cincinnati-Milacron T[3] Wrist

Joint i	α_i	a_i	d_i	$\theta_{j,i}$
1	α	0	d_1	$\theta_{2,1}$ (variable)
2	$-\alpha$	0	0	$\theta_{3,2}$ (variable)
3	0	0	d_3	$\theta_{4,3}$ (variable)

Hartenberg method or the method of successive screw displacements. Both methods have been discussed in detail in Chapter 2.

7.7.2 Transformation between Joint Space and Actuator Space

The second step is achieved by applying the theory of fundamental circuits and the coaxiality conditions. With the aid of graph representation, the fundamental circuits are identified and the fundamental circuit equation, Eq. (7.7), is written once for each fundamental circuit. Next, the coaxiality condition, Eq. (7.9), is written as many times as appropriate for the purpose of eliminating some of the unwanted angular displacements. This leads to a simple relationship between the actuator displacements and the joint angles of the equivalent open-loop chain.

For example, the Cincinnati-Milacron T^3 wrist shown in Fig. 7.7 is a 3-dof mechanism with three bevel gear pairs. Applying the structure characteristic no. 4, three fundamental circuits are identified as 7–2–3–4–7, 6–1–2–7–6, and 5–1–2–3–5. Applying the structure characteristic no. 8, we obtain vertices 3, 2, and 2 as the corresponding transfer vertices. Hence the fundamental circuit equations are given by

$$f(7,4,3): \quad \theta_{7,3} = -N_{47}\theta_{4,3}, \tag{7.28}$$

$$f(6,7,2): \quad \theta_{6,2} = -N_{76}\theta_{7,2}, \tag{7.29}$$

$$f(5,3,2): \quad \theta_{5,2} = N_{35}\theta_{3,2}. \tag{7.30}$$

Since link 1 is considered as the fixed link, all angular displacements should be referred to link 1. The input angular displacements are $\theta_{2,1}$, $\theta_{5,1}$, and $\theta_{6,1}$, while the joint angles of the equivalent open-loop chain are $\theta_{2,1}$, $\theta_{3,2}$, and $\theta_{4,3}$. All the other angular displacements should be eliminated by applying the coaxial conditions. From Fig. 7.7 we observe that links 2, 3, and 7 share a common joint axis, z_2, and links 1, 2, 5, and 6 share another common joint axis, z_1. Hence three coaxiality conditions can be written as

$$\theta_{7,3} = \theta_{7,2} - \theta_{3,2}, \tag{7.31}$$

$$\theta_{6,2} = \theta_{6,1} - \theta_{2,1}, \tag{7.32}$$

$$\theta_{5,2} = \theta_{5,1} - \theta_{2,1}. \tag{7.33}$$

Solving Eqs. (7.28) through (7.33) for $\theta_{5,1}$ and $\theta_{6,1}$ in terms of $\theta_{2,1}$, $\theta_{3,2}$, and $\theta_{4,3}$ yields

$$\theta_{5,1} = \theta_{2,1} + N_{35}\theta_{3,2}, \tag{7.34}$$

$$\theta_{6,1} = \theta_{2,1} - N_{76}\theta_{3,2} + N_{76}N_{47}\theta_{4,3}. \tag{7.35}$$

Since link 2 is an input link, we have

$$\theta_{2,1} = \theta_{2,1}. \tag{7.36}$$

We can write Eqs. (7.34), (7.35), and (7.36) in matrix form as

$$\begin{bmatrix} \theta_{2,1} \\ \theta_{5,1} \\ \theta_{6,1} \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 1 & N_{35} & 0 \\ 1 & -N_{76} & N_{76}N_{47} \end{bmatrix} \begin{bmatrix} \theta_{2,1} \\ \theta_{3,2} \\ \theta_{4,3} \end{bmatrix}, \tag{7.37}$$

or simply

$$\boldsymbol{\phi} = A\boldsymbol{\theta}, \tag{7.38}$$

where $\boldsymbol{\phi} = [\theta_{2,1}, \theta_{5,1}, \theta_{6,1}]^{\mathrm{T}}$ denotes the vector of input angular displacements, $\boldsymbol{\theta} = [\theta_{2,1}, \theta_{3,2}, \theta_{4,3}]^{\mathrm{T}}$ denotes the vector of joint angles, and

$$A = \begin{bmatrix} 1 & 0 & 0 \\ 1 & N_{35} & 0 \\ 1 & -N_{76} & N_{76}N_{47} \end{bmatrix}. \tag{7.39}$$

For the direct kinematics, the input angular displacements are given and the joint angles are found from inverse transformation of Eq. (7.38). Once the joint angles are known, the end-effector orientation is found by the direct Denavit–Hartenberg transformation. For the inverse kinematics, the end-effector orientation is given. We first find the joint angles from the inverse Denavit–Hartenberg transformation. Then we find the input angular displacements from Eq. (7.38).

7.7.3 Angular Velocity Relations

The angular velocity of the end effector with respect to the base link, $\boldsymbol{\omega}_{4,1}$, can be expressed as a summation of the instantaneous twists about the three equivalent joint axes:

$$\boldsymbol{\omega}_{4,3} = \dot{\theta}_{2,1}\mathbf{z}_1 + \dot{\theta}_{3,2}\mathbf{z}_2 + \dot{\theta}_{4,3}\mathbf{z}_3, \tag{7.40}$$

where $\mathbf{z}_i$ denotes a unit vector pointing along the positive z_i-axis. Taking the time derivative of Eq. (7.37) yields

$$\begin{bmatrix} \dot{\theta}_{2,1} \\ \dot{\theta}_{5,1} \\ \dot{\theta}_{6,1} \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 1 & N_{35} & 0 \\ 1 & -N_{76} & N_{76}N_{47} \end{bmatrix} \begin{bmatrix} \dot{\theta}_{2,1} \\ \dot{\theta}_{3,2} \\ \dot{\theta}_{4,3} \end{bmatrix}. \tag{7.41}$$

Hence the joint rates can be expressed in terms of the actuator rotation rates by taking the inverse transformation of Eq. (7.41); that is,

$$\begin{bmatrix} \dot{\theta}_{2,1} \\ \dot{\theta}_{3,2} \\ \dot{\theta}_{4,3} \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ -N_{53} & N_{53} & 0 \\ -N_{74}(N_{67}+N_{53}) & N_{74}N_{53} & N_{74}N_{67} \end{bmatrix} \begin{bmatrix} \dot{\theta}_{2,1} \\ \dot{\theta}_{5,1} \\ \dot{\theta}_{6,1} \end{bmatrix}. \tag{7.42}$$

For the direct velocity problem, the actuator rotation rates are given. We compute the joint rates from Eq. (7.42) and the angular velocity of the end effector from Eq. (7.40). For the inverse velocity problem, the angular velocity of the end effector is given. We solve the joint rates from the vector equation (7.40) and then the actuator rotation rates from Eq. (7.41).

7.8 STATIC FORCE ANALYSIS

In this section we apply the principle of virtual work to derive a transformation between the equivalent joint torques and the actuator torques.

7.8.1 Principle of Virtual Work

For a 3-dof wrist, the virtual displacements of the input actuators can be written as $\delta\boldsymbol{\phi} = [\delta\phi_1, \delta\phi_2, \delta\phi_3]^{\mathrm{T}}$, and the virtual displacements at the joints of the equivalent open-loop chain can be written as $\delta\boldsymbol{\theta} = [\delta\theta_{2,1}, \delta\theta_{3,2}, \delta\theta_{4,3}]^{\mathrm{T}}$. Let the actuator input torques be denoted by $\boldsymbol{\xi} = [\xi_1, \xi_2, \xi_3]^{\mathrm{T}}$ and the output joint torques be denoted by $\boldsymbol{\tau} = [\tau_1, \tau_2, \tau_3]^{\mathrm{T}}$, respectively. Neglecting the frictional forces and the gravitational effect, the virtual work, δW, produced by these active forces is given by

$$\delta W = \boldsymbol{\xi}^{\mathrm{T}}\,\delta\boldsymbol{\phi} - \boldsymbol{\tau}^{\mathrm{T}}\,\delta\boldsymbol{\theta}. \tag{7.43}$$

From Eq. (7.38) we obtain a relation between the virtual displacements $\delta\boldsymbol{\phi}$ and $\delta\boldsymbol{\theta}$ as

$$\delta\boldsymbol{\phi} = A\,\delta\boldsymbol{\theta}. \tag{7.44}$$

Substituting Eq. (7.44) into (7.43) yields

$$\delta W = (\boldsymbol{\xi}^{\mathrm{T}} A - \boldsymbol{\tau}^{\mathrm{T}})\,\delta\boldsymbol{\theta}. \tag{7.45}$$

The system is under equilibrium if and only if the virtual work vanishes for any independent infinitesimal virtual displacement. Hence

$$\boldsymbol{\xi}^{\mathrm{T}} A - \boldsymbol{\tau}^{\mathrm{T}} = 0. \tag{7.46}$$

Taking the transpose of Eq. (7.46) yields

$$\boldsymbol{\tau} = B\boldsymbol{\xi}, \tag{7.47}$$

where $B = A^{\mathrm{T}}$ is called the *structure matrix*. For example, the structure matrix for the Cincinnati-Milacron T^3 wrist is given by

$$B = A^{\mathrm{T}} = \begin{bmatrix} 1 & 1 & 1 \\ 0 & N_{35} & -N_{76} \\ 0 & 0 & N_{76}N_{47} \end{bmatrix}.$$

We note that the structure matrix is a function of the structure topology and the gear ratios and is independent of end-effector orientation.

For a 3-dof spherical wrist, it can be shown that the vector of equivalent joint torques is related to the end-effector output torque, $\mathbf{n} = [n_x, n_y, n_z]^{\mathrm{T}}$, by

$$\boldsymbol{\tau} = J^{\mathrm{T}}\mathbf{n}, \tag{7.48}$$

where J is a 3×3 Jacobian matrix relating the end-effector angular velocity to the joint rates of the equivalent open-loop chain. Combining Eqs. (7.47) and (7.48), we obtain an overall transformation as

$$\boldsymbol{\xi} = B^{-1}J^{\mathrm{T}}\mathbf{n}. \tag{7.49}$$

Using Eq. (7.49), we can compute the input actuator torques for any desired end-effector output torque. On the other hand, given the input actuator torques, we can compute the resulting end-effector torque by performing the inverse transformation of Eq. (7.49).

7.8.2 Transmission Lines

In this section we introduce the concept of *transmission line*. Taking the derivative of Eq. (7.47) yields

$$d\boldsymbol{\tau} = B\,d\boldsymbol{\xi}. \tag{7.50}$$

Hence the (i, j) element of the structure matrix B can be interpreted as the partial rate of change of the joint torque with respect to the input actuator torque; that is,

$$b_{ij} = \frac{\partial \tau_i}{\partial \xi_j}, \tag{7.51}$$

where b_{ij} denotes the (i, j) element of B.

The element $b_{ij} \neq 0$ implies that the input torque ξ_j is amplified by b_{ij} times when it is transmitted to the ith joint. Hence $b_{ij} = 0$ implies that the input torque ξ_j does not have any influence on the resultant torque at joint i. The ith row of the structure matrix B describes how the resultant torque about joint i is affected by all the input actuators. On the other hand, the jth column of the structure matrix B describes how the torque of the jth input actuator is transmitted to various joints of a mechanism. Since input torques are transmitted to the joints by gear trains, nonzero elements in a column of the structure matrix must be consecutive. The gear train that results in a series of nonzero elements in the jth column of B is therefore called a *transmission line* for the input actuator j. In a transmission line, the first gear is attached to the rotor of an actuator, and the last gear is attached to one of the links in the equivalent open-loop chain.

Following the discussion above, it can be concluded that the element b_{ij} is equal to the *train value* defined from the input gear j to the gear pivoted about the ith joint axis of the equivalent open-loop chain. For example, the Cincinnati-Milacron T^3 wrist shown in Fig. 7.7 has three transmission lines: (1) link 2 by itself, (2) gears 5 and 3, and (3) gears 6, 7, 7′, and 4. The element b_{23} is equal to the train value $-N_{76}$ defined from the input gear 6 to gear 7 pivoted on the second joint axis, where the negative sign comes from the fact that when the input gear 6 makes a positive rotation about the z_1-axis, gear 7 will make a negative rotation about the z_2-axis. Similarly, the element b_{33} is equal to the train value $N_{76}N_{47}$ defined from the input gear 6 to gear 4 pivoted about the third joint axis. Hence the elements of B can be determined by inspection. Note that if the stator of a motor is mounted on the ith link and the rotor is connected to the $(i+1)$th link without any gear reduction, we have a direct drive. The element of B for a direct drive is equal to 1.

7.8.3 Kinematics of the Bendix Wrist

In this section the kinematics of the Bendix wrist shown in Fig. 7.4 is analyzed. The canonical graph representation of the mechanism is shown in Fig. 7.14, where vertex 1 represents the base link and 4 represents the end effector. The thin-edged path, which starts from the base link and ends at the end-effector link, consists of vertices 1, 2, 3, and 4. Hence the equivalent open-loop chain is made up of links 1, 2, 3, and 4 connected in series by revolute joints as shown in Fig. 7.15. The transformation between the end-effector space and the joint space is similar to that for the Cincinnati-Milacron T^3 wrist. The only difference is in the twist angles between the joint axes. In the Bendix design the twist angles are $\alpha_1 = 90°$, $\alpha_2 = -90°$, and $\alpha_3 = 0°$.

There are four gear pairs and therefore four fundamental circuits. Applying the structure characteristic no. 4, we identify four fundamental circuits as 5–

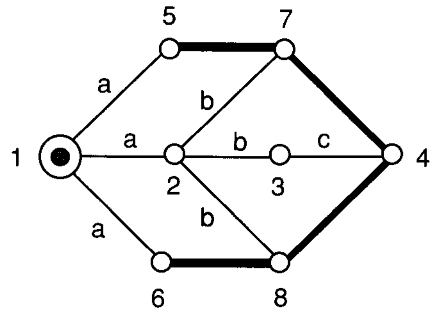


FIGURE 7.14. Canonical graph representation of the Bendix wrist.

1–2–7–5, 7–2–3–4–7, 8–2–3–4–8, and 6–1–2–8–6. Applying the structure characteristic no. 8, we obtain vertices 2, 3, 3, and 2 as the corresponding transfer vertices. Hence the fundamental circuit equations are given by

$$f(7,5,2): \quad \theta_{7,2} = N_{57}\theta_{5,2}, \tag{7.52}$$

$$f(4,7,3): \quad \theta_{4,3} = -N_{74}\theta_{7,3}, \tag{7.53}$$

$$f(4,8,3): \quad \theta_{4,3} = N_{84}\theta_{8,3}, \tag{7.54}$$

$$f(8,6,2): \quad \theta_{8,2} = -N_{68}\theta_{6,2}. \tag{7.55}$$

Since link 1 is considered the fixed link, all angular displacements should be referred to link 1. The input angular displacements are $\theta_{2,1}$, $\theta_{5,1}$, and $\theta_{6,1}$,

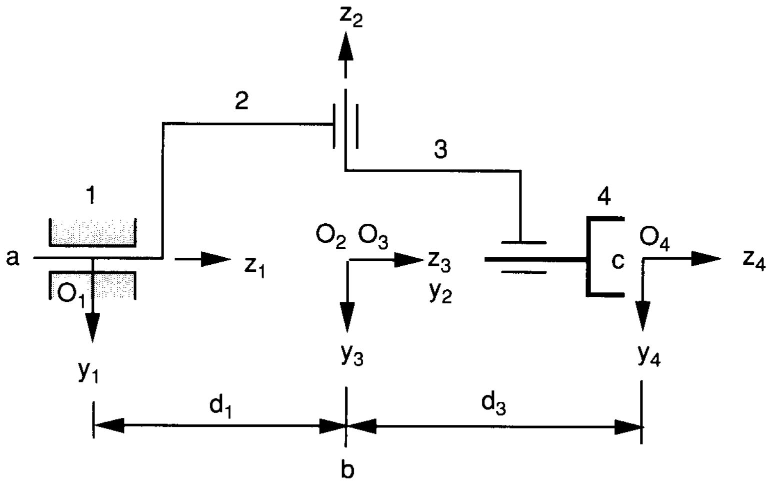


FIGURE 7.15. Equivalent open-loop chain of the Bendix wrist.

while the joint angles of the equivalent open-loop chain are $\theta_{2,1}$, $\theta_{3,2}$, and $\theta_{4,3}$. All the other angular displacements should be eliminated by applying the coaxial conditions.

From Fig. 7.4 we observe that links 2, 3, 7, and 8 share one common joint axis, z_2, and links 1, 2, 5, and 6 share another common joint axis, z_1. Hence four coaxiality conditions can be written as

$$\theta_{7,3} = \theta_{7,2} - \theta_{3,2}, \tag{7.56}$$

$$\theta_{8,3} = \theta_{8,2} - \theta_{3,2}, \tag{7.57}$$

$$\theta_{5,2} = \theta_{5,1} - \theta_{2,1}, \tag{7.58}$$

$$\theta_{6,2} = \theta_{6,1} - \theta_{2,1}. \tag{7.59}$$

Solving Eqs. (7.52) through (7.59) for $\theta_{5,1}$ and $\theta_{6,1}$ in terms of $\theta_{2,1}$, $\theta_{3,2}$, and $\theta_{4,3}$, we obtain

$$\theta_{5,1} = \theta_{2,1} + N_{75}\theta_{3,2} - N_{75}N_{47}\theta_{4,3}, \tag{7.60}$$

$$\theta_{6,1} = \theta_{21} - N_{86}\theta_{3,2} - N_{86}N_{48}\theta_{4,3}. \tag{7.61}$$

Since link 2 is an input link, we have

$$\theta_{2,1} = \theta_{2,1}. \tag{7.62}$$

Writing Eqs. (7.60), (7.61), and (7.62) in matrix form, we obtain

$$\begin{bmatrix} \theta_{2,1} \\ \theta_{5,1} \\ \theta_{6,1} \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 1 & N_{75} & -N_{75}N_{47} \\ 1 & -N_{86} & -N_{86}N_{48} \end{bmatrix} \begin{bmatrix} \theta_{2,1} \\ \theta_{3,2} \\ \theta_{4,3} \end{bmatrix}. \tag{7.63}$$

Hence the structure matrix B is given by

$$B = A^{\mathrm{T}} = \begin{bmatrix} 1 & 1 & 1 \\ 0 & N_{75} & -N_{86} \\ 0 & -N_{75}N_{47} & -N_{86}N_{48} \end{bmatrix}. \tag{7.64}$$

In view of the right angle between the z_1 and z_2 axes, and between the z_2 and z_3 axes, we have $N_{75} = N_{86} = 1/N_{47} = 1/N_{48}$. Hence the structure matrix reduces further to

$$B = A^{\mathrm{T}} = \begin{bmatrix} 1 & 1 & 1 \\ 0 & N_{75} & -N_{75} \\ 0 & -1 & -1 \end{bmatrix}. \tag{7.65}$$

Once the structure matrix is known, the velocity and statics analyses can be performed by following the procedure outlined earlier for the Cincinnati-Milacron wrist.

7.9 SUMMARY

The graph representation of the kinematic structures of epicyclic gear trains, including bevel-gear wrist mechanisms, has been described. Applying the network properties of graphs, the structure characteristics of epicyclic gear trains were derived and bevel-gear wrist mechanisms classified according to their structure characteristics. The theory of fundamental circuit and the coaxiality condition were introduced for the kinematic analysis of epicyclic gear trains. It was shown that the kinematics of bevel-gear wrist mechanisms can be analyzed in two basic steps. The first step involves the identification of an equivalent open-loop chain and the derivation of the kinematic relationship between the orientation of the end effector and the joint angles of the equivalent open-loop chain. The second step deals with the kinematic relationship between the joint angles and the input actuator displacements. The concepts of transmission lines and structure matrix were also introduced. It was shown that using the concept of transmission lines, the structure matrix of a wrist mechanism can be derived by an inspection of the kinematic structure of a mechanism. Several industrial gear reducers and wrist mechanisms were analyzed to illustrate the methodology.

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EXERCISES

1. Sketch the graph and canonical graph of the mechanism shown in Fig. 7.16. Calculate the degrees of freedom of this mechanism.

2. Sketch the graph and canonical graph of the wrist mechanism shown in Fig. 7.17. How many degrees of freedom does this mechanism have? What links can be used as the input links, and why?

3. Discuss the advantages and disadvantages associated with simple and oblique wrist mechanisms from the kinematic structural point of view.

4. Find the speed reduction ratio ω_1/ω_2 for the compound planetary gear train shown in Fig. 7.18, where link 1 is the input link and link 2 is the output link. Find a set of gears that will provide a reduction ratio of approximately $2:1$.

5. Find the speed reduction ratio ω_3/ω_4 for the planetary gear train shown in Fig. 7.19, where link 3 is the input link and link 4 is the output link. Find a set of gears that will provide a reduction ratio of approximately $3:1$.

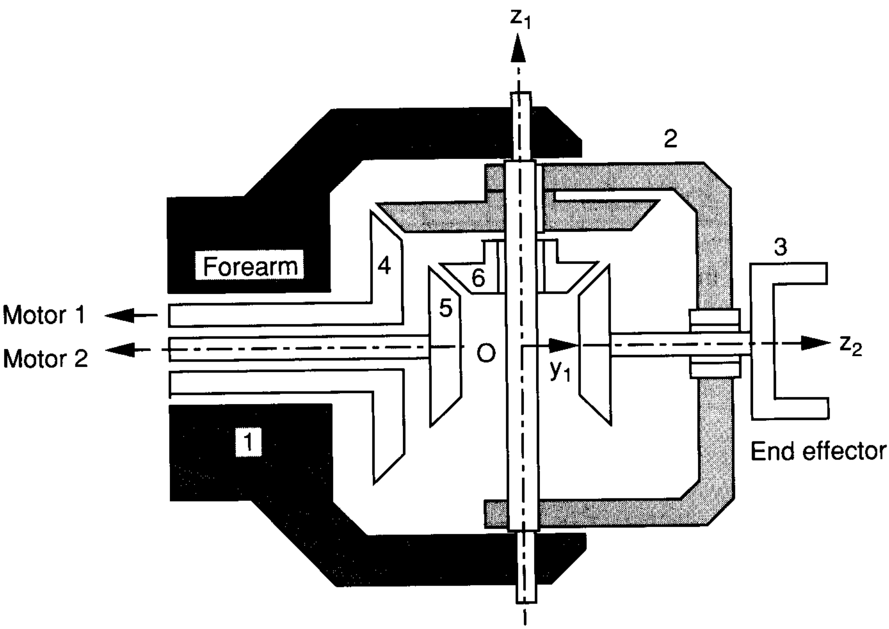

FIGURE 7.16. Spherical wrist mechanism—1.

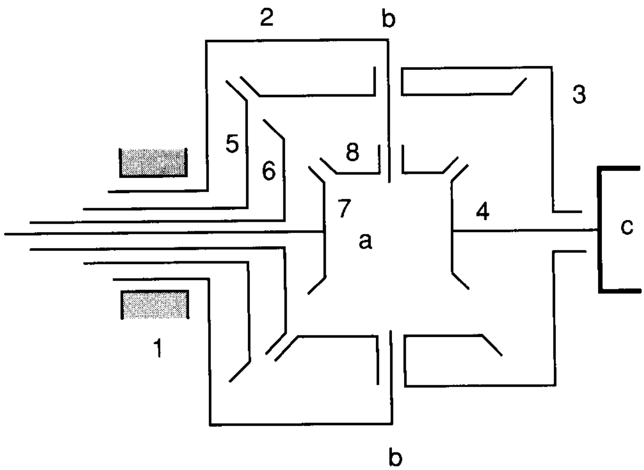

FIGURE 7.17. Spherical wrist mechanism—2.

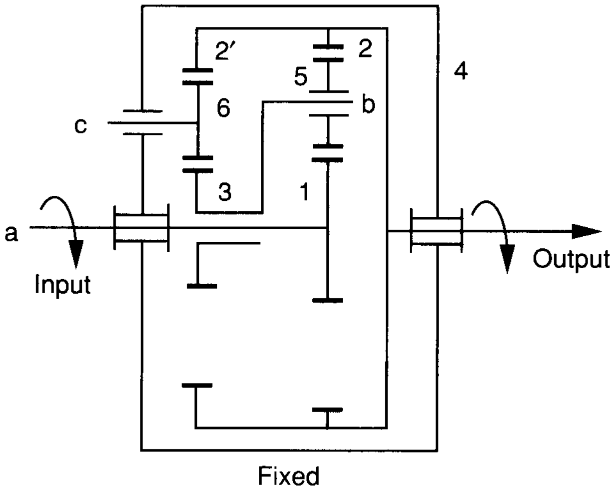

FIGURE 7.18. Compound planetary gear train—1.

6. Derive the structure matrix for the 3-dof spherical wrist mechanism shown in Fig. 7.20, where links 6, 9, and 10 are the input links and link 4 is the end effector.

7. Derive the structure matrix for the 3-dof spherical wrist mechanism shown in Fig. 7.21, where links 6, 9, and 10 are the input links and link 4 is the end effector.

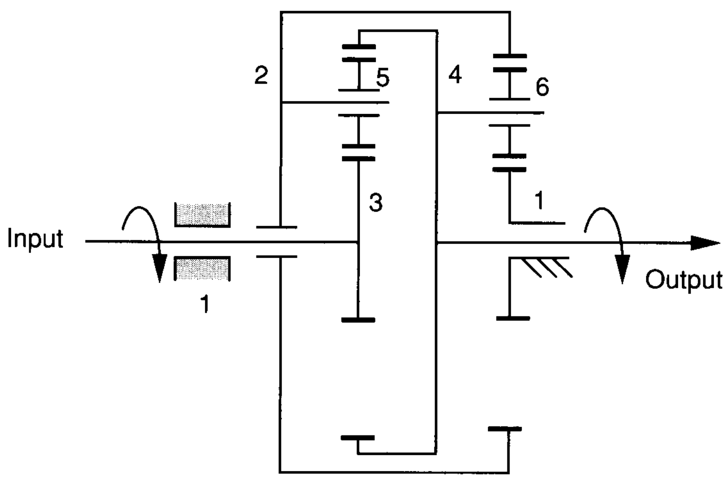

FIGURE 7.19. Compound planetary gear train—2.

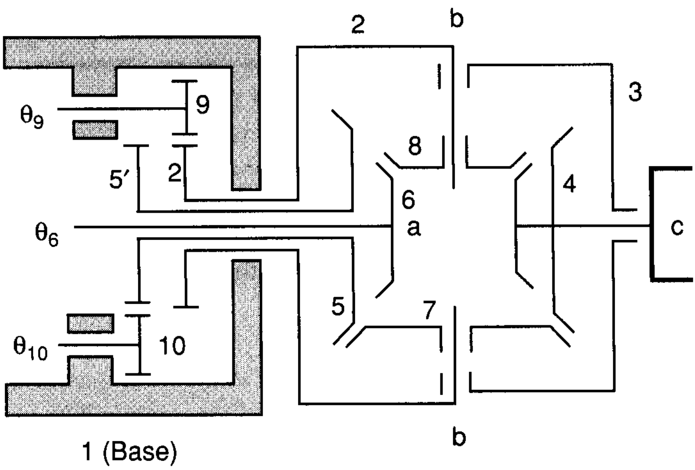

FIGURE 7.20. Spherical wrist mechanism—3.

8. For the spherical wrist mechanism shown in Fig. 7.17, let links 5, 6, and 7 be the input links and 4 be the end-effector link. Find the input rotation rates in terms of a given angular velocity and general posture of the end effector.

9. For the spherical wrist mechanism shown in Fig. 7.16, links 4 and 5 are the input links and 3 is the end-effector link. Assuming that the end effector is oriented at a general posture, calculate the input torques, τ_4 and τ_5,

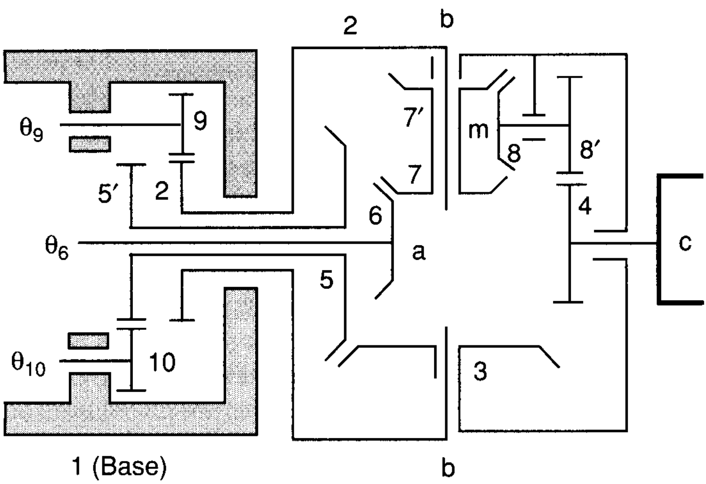

FIGURE 7.21. Spherical wrist mechanism—4.

required to produce an end-effector ouput moment of (n_x, n_y, n_z) about the center of the wrist.

10. For the 3-dof spherical wrist mechanism shown in Fig. 7.22, links 7, 9, and 10 are the input links and link 4 is the end effector. Applying the principle of virtual work, calculate the input torques, τ_7, τ_9 and τ_{10}, required to generate an end-effector output moment of (n_x, n_y, n_z) about the center of the wrist. For the particular posture shown in Fig. 7.22, is it possible to generate an output moment in the x_1-direction?

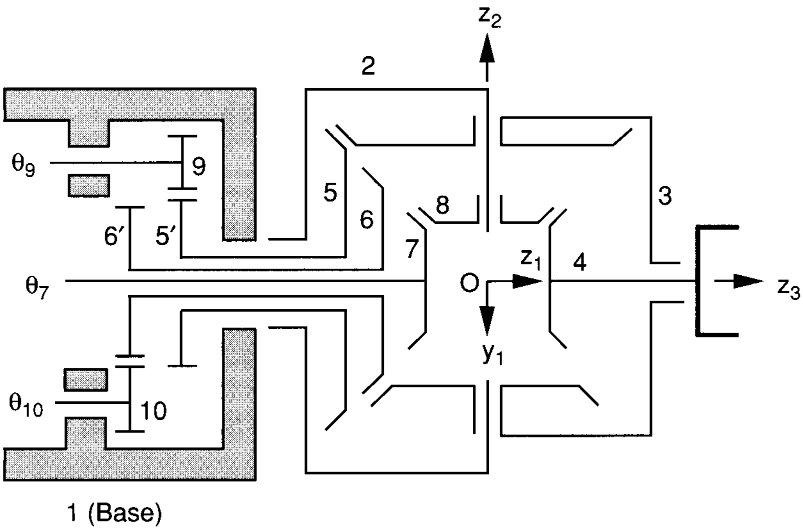

FIGURE 7.22. Spherical wrist mechanism—5.